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Synthesis, antifungal evaluation and optical properties of silver molybdate microcrystals in different solvents: a combined experimental and theoretical study†

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In this study, we investigate the structure, antifungal activity, and optical properties of β-Ag_2MoO_4 using experimental and theoretical approaches. β-Ag_2MoO_4 samples were prepared by a co-precipitation method using different solvents (water, ethanol and ammonia), and their antifungal activity against *Candida albicans* was investigated. The samples were characterized by X-ray diffraction, micro-Raman spectroscopy, field emission scanning electron microscopy, and transmission electron microscopy with energy dispersive spectroscopy. The optical properties were investigated by UV-Vis spectroscopy and photoluminescence measurements at room temperature. The thermodynamic equilibrium shape of the β-Ag_2MoO_4 crystals was determined based on the surface energies calculated using Wulff construction. The (011) orientation was the predominant surface in the morphology. The experimental morphology was obtained by varying the surface energy ratio for each facet. A large decrease in surface energy for the (111) surface provided the experimental morphology for crystals synthesized using water and ethanol as solvents; when the surface energies for both (011) and (001) surfaces increased, the crystal morphology obtained using ammonia as a solvent was reproduced. A correlation between the exposed surfaces and antifungal activity was revealed, and an explanation to this behavior that arises from different morphologies and structural data was provided. Theoretical calculations confirm the rationality of the experimental scheme and elucidate the underlying reason for the fungistatic and fungicidal activity against *Candida albicans*.

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1. Introduction

Metal molybdates are an important family of inorganic materials, and have both technological and theoretical importance. Indeed, they exhibit a wide range of electrical and optical properties that can find widespread applications in various fields such as photocatalysis, solar energy conversion, energy storage, as well as antibacterial agents.[1–10]

Silver molybdate (Ag_2MoO_4) is an important member of this family and exists at ambient pressure in two allotropic forms: a metastable α-tetragonal form (obtained at pH 5–6), which is irreversibly transformed to a β-Ag_2MoO_4 (above 280 °C,[11] with a cubic spinel structure).[11–13] β-Ag_2MoO_4 has been intensively investigated using various techniques: Arora *et al.* studied its behavior at high pressure using X-ray diffraction (XRD) and Raman spectroscopy,[14] while a first-principles study of the pressure-induced phase transition and electronic properties was recently reported.[15] The various properties of β-Ag_2MoO_4 enable a number of practical applications, such as surface enhanced Raman scattering techniques,[16] fast ion-conducting glasses when associated with AgI,[17] high temperature lubrication,[18,19] and enhanced photoluminescence after association with silver tungstate in nanocomposites.[20] On the other hand,

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the quest to design and fabricate new antibacterial surfaces is an important task to meet the urgent demands of biomedical applications.

Different synthetic approaches have been described in order to obtain Ag_2MoO_4 ceramics, microcrystals, and nanoparticles of various shapes, depending on the intended application. Various β-Ag_2MoO_4 nano- or microstructures have been fabricated, such as nanoparticles,[21–23] wire-like nanostructures,[24,25] flower-like microstructures,[26] and well-shaped microcrystals[20,27] by different processes. However, the most common preparation technique is the hydrothermal method, frequently used in the synthesis of inorganic semiconducting nanostructures (see the review by H. Zhang *et al.*[27]). Indeed, the mild conditions of this technique enable significant advantages, namely, relatively low processing temperature and a short reaction time. In a recent study, we reported the preparation and characterization of β-Ag_2MoO_4 microcrystals using a microwave-assisted hydrothermal (MAH) method.[28] However, it has been well established that their morphology is extremely sensitive to experimental conditions such as pH, reaction time, temperature, and pressure,[29,30] resulting in potentially different behaviors.

This study combines both experimental and theoretical approaches. β-Ag_2MoO_4 crystals were prepared by a co-precipitation method (CP) using water, ethanol, and ammonia as solvents. The structural characterization of the samples was performed using XRD, and micro-Raman (MR) spectroscopy; the morphology, shape, and elemental composition of the crystals were studied by field emission scanning electron microscopy (FE-SEM), transmission electron microscopy (TEM), high-resolution TEM, and energy dispersive X-ray spectrometry (EDS); their optical properties were investigated by UV–Vis absorption spectroscopy, and photoluminescence (PL) measurements at room temperature. In addition, their fungistatic and fungicidal activity against *Candida albicans* was also investigated, and to gain a better understanding of these properties, in this paper, we propose a model to elucidate the fungicidal mechanism of β-Ag_2MoO_4. To complement the experimental results, first-principles calculations, based on density functional theory (DFT), were performed in order to provide a deeper understanding of the atomic and electronic structure, and reveal a correlation among morphology, optical properties, and antifungal activities.

This paper is organized as follows: section 2 describes the experimental and computational details; the experimental and theoretical results are presented and discussed in detail in section 3; and a brief summary and the main conclusions are presented in section 4.

2. Experimental

2.1 Synthesis of β-Ag_2MoO_4 using different solvents

β-Ag_2MoO_4 microcrystals were synthesized by a surfactant-free CP method using water, ethanol, and ammonia as solvents. Initially, 1 mmol of sodium molybdate dihydrate ($Na_2MoO_4{\cdot}2H_2O$; 99.5% purity, Sigma-Aldrich) and 2 mmol of silver nitrate ($AgNO_3$; 99.8% purity, Sigma-Aldrich) were separately dissolved (solutions 1 and 2, respectively) in 50 mL of deionized water. Solution 2 was then added dropwise to solution 1 under vigorous magnetic stirring, and heated at 90 °C for 10 min. Then, the same procedure as previously described was employed for the synthesis of β-Ag_2MoO_4 using ethanol as the solvent. Under vigorous magnetic stirring, solution 2 was slowly added dropwise to solution 1, and this mixture was heated at 70 °C for 10 min. Using ammonia as the solvent, 1 mmol of sodium molybdate dihydrate was dissolved in 47 mL of deionized water to prepare solution 1; then, 2 mmol of silver nitrate were dissolved in 50 mL of deionized water (solution 2). Solution 2 was then added dropwise to solution 1, and 3 mL of an ammonium hydroxide solution (34% Sigma-Aldrich) was added dropwise into this mixture. This solution was stirred and heated at 90 °C for 10 min. The precipitates obtained using different solvents were collected by centrifugation, washed several times with deionized water and ethanol until pH = 7, and then dried at 60 °C for 12 h.

2.2 Characterization

The structure of β-Ag_2MoO_4 microcrystals was assessed by XRD using a D/Max-2500PC diffractometer (Rigaku) with Cu Kα radiation (λ = 1.5406 Å) in the 2θ range between 10° and 70°, with a scanning rate and step size of 2° min^{-1} and 0.02°, respectively. MR measurements were recorded using a modular Raman spectrometer (Horiba, Jobin Yvon), model RMS-550 with an Ar laser excitation of 514 nm, and a fiber-microscope operating at 30–1000 cm^{-1}. The morphology and size of the β-Ag_2MoO_4 structures were determined by FE-SEM (Inspect F50, FEI Company Hillsboro), operating at 5 kV. UV-Vis spectra were recorded using a Varian spectrophotometer (Model Cary 5G) in the diffuse reflectance mode. PL spectra were collected with a Thermal Jarrel Ash Monospec monochromator and a Hamamatsu R446 Photomultiplier. The 350.7 nm (2.57 eV) excitation wavelength of a Kr-ion laser (Coherent Innova) was used, with the laser output maintained at 200 mW. All measurements were carried out at room temperature.

2.3 Antifungal effect of β-Ag_2MoO_4 microcrystals

The antifungal activity of β-Ag_2MoO_4 microcrystals was evaluated by determining the minimum inhibitory concentration (MIC) and minimum fungicidal concentration (MFC) on planktonic cells. The tests were performed according to the Clinical and Laboratory Standards Institute (CLSI) broth microdilution method, document M27-A3.[31] A standard strain of *Candida albicans* from the American Type Culture Collection (ATCC 90028) was streaked onto a Sabouraud dextrose agar supplemented with chloramphenicol (0.05 g L^{-1}) (SDA, Acumedia Manufacturers Inc., Baltimore, MD, USA), and incubated for 48 h at 37 °C. One loopful of this culture was then transferred to RPMI-1640 (Sigma-Aldrich). After 21 h of incubation in an orbital shaker (75 rpm) at 37 °C, the cells were washed twice with a phosphate-buffered saline solution (PBS, pH 7.2)

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using a centrifuge at 5000*g* for 5 min (rotor model A462). The washed fungal cells were resuspended in a RPMI 1640 culture medium to 10^7 colony-forming units per milliliter (CFU mL^{-1}) by adjusting the optical density of the suspension to 0.6 at 540 nm. The inoculum was then serially diluted according to the CLSI protocol, and the final concentration of *C. albicans* was 10^4 CFU mL^{-1}. To determine MIC and MFC parameters, *C. albicans* was incubated in 96-well microtiter plates for 48 h at 35 °C, and exposed to two-fold serial dilutions of each of the β-Ag_2MoO_4 solutions (from 1000 to 1.95 μg mL^{-1}) in RPMI-1640. An inoculated RPMI-1640 medium without β-Ag_2MoO_4 microcrystals was used as positive control, while an uninoculated RPMI-1640 medium was used as negative control. The MICs were defined as the lowest concentrations of each solution required for complete growth inhibition (no visible growth by visual inspection). To determine the MFC values, aliquots from 10-fold serial dilutions (10^{-1} to 10^{-4}) of each well were removed and inoculated (25 μL) on SDA plates in duplicate. After 48 h of incubation at 37 °C, the CFU mL^{-1} were calculated and log_{10} transformed. The MFC values were defined as the lowest concentration for each of the β-Ag_2MoO_4 solutions resulting in no fungal growth on the plates. The assays were performed in triplicate on three separate occasions.

2.4 Model system and computational method

First-principles total-energy calculations were carried out within the DFT framework using the Vienna *ab initio* simulation package (VASP).[32,33] The Kohn–Sham equations were solved using the Perdew–Burke–Ernzerhof exchange–correlation functional,[34,35] and the conjugated gradient energy minimization method was used to obtain relaxed systems until the residual forces experienced by each atom were smaller than 0.01 eV $Å^{-1}$. The projector augmented wave (PAW) method combined with the frozen core approximation was used for the ion–electron interactions.[36,37] The plane-wave expansion was truncated at a cut-off energy of 460 eV, and the Brillouin zones were sampled using Monkhorst–Pack special *k*-point grids that guarantee geometrical and energetic convergence for the Ag_2MoO_4 structures. E_{surf} is defined as the total energy per repeating cell of the slab (E_{slab}) minus the total energy of the perfect crystal per molecular unit (E_{bulk}) multiplied by the number of molecular units of the surface (n), divided by the surface area per repeating cell of the two sides of the slab, as shown in eqn (1). In order to confirm the convergence of the total energy with respect to the slab thickness of different surface models, E_{surf} for several low-index planes was calculated. The equilibrium shape of a crystal can be calculated by the classic Wulff construction that minimizes the total surface free energy at a fixed volume, providing a simple correlation between the surface energy (E_{surf}) of the (hkl) plane and its distance *rhkl* in the normal direction from the center of the crystallite.

$$E_{surf} = \frac{E_{slab} - nE_{bulk}}{2A} \quad (1)$$

The conventional cubic unit cell of β-Ag_2MoO_4 contained eight formula units. The Mo atoms occupy 8a tetrahedral sites, the Ag atoms reside at the octahedral 16d position, and the O atoms remain at 32e positions. The (001), (011) and (111) surfaces of β-Ag_2MoO_4 were modeled by an unreconstructed slab model using a calculated equilibrium geometry (a = 9.454 Å, and u(O) = 0.2351) using a (3 × 3 × 1) Monkhorst–Pack special *k*-point grid. A vacuum spacer of 15 Å was introduced in the *z*-direction to prevent surfaces from interacting with each other.

3. Results and discussion

3.1 X-ray diffraction

XRD patterns of β-Ag_2MoO_4 synthesized using different solvents are shown in Fig. 1. All XRD peaks of the β-Ag_2MoO_4 crystals indicate a cubic spinel structure with a space group ($Fd\bar{3}m$) and point-group symmetry (O_{7h}) without any deleterious phases, in good agreement with the inorganic crystal structure database (ICSD) card no. 28891.[13] These microcrystals presented sharp and well-defined diffraction peaks, which indicate a long-range structural order and crystallinity. The experimental results confirm the formation of β-Ag_2MoO_4, and are in good agreement with previous reports showing the formation of a cubic spinel phase when the pH was near seven,[11,26,30] the same pH conditions that were used in this study.

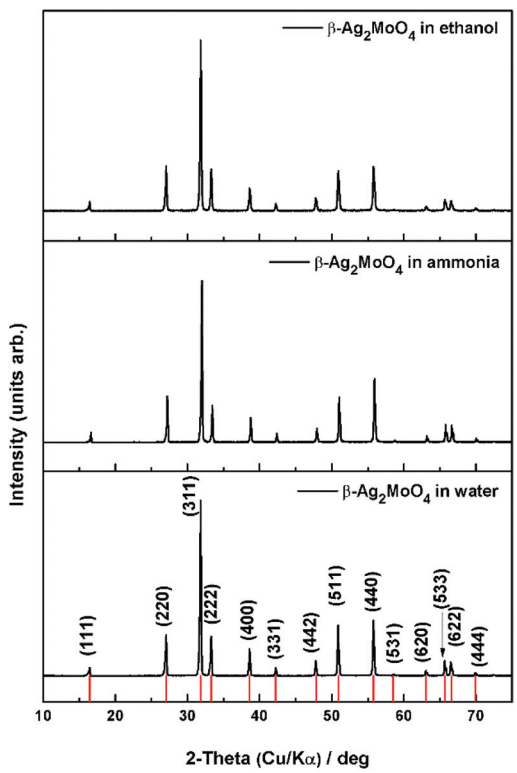

Fig. 1 XRD patterns of the β-Ag_2MoO_4 microcrystals obtained using different solvents. Vertical red lines indicate the position and relative intensity of the corresponding ICSD card no. 28891.

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3.2 FE-SEM analysis

The microstructure of the β-Ag_2MoO_4 microcrystals was studied by FE-SEM. The particles were round-shaped, typically in the micrometer size. This microstructure contrasts strongly with that recently reported for MAH samples,[27] where microcrystals with a small size and homogeneous shape (cube-like) were only obtained when the processing temperature increased to 160 °C. Interestingly, round nanoparticles, similar to our CP samples, were reported using a surfactant-free MAH method as well-formed clusters in the presence of polyvinylpyrrolidone at a similar temperature of 180 °C.[28] The growth of β-Ag_2MoO_4 nanowires at a pH of 4 (ref. 20) and even 2 (ref. 25) was reported, while other authors observed that, in this pH range, di- and/or tri $Ag_2Mo_2O_7$ and $Ag_2Mo_3O_{10}$ samples were obtained.[26,30]

Fig. 2(a–c) show the micrographs of ~1 μm isolated microcrystals after irradiation exposure with an electron beam operated at 5 kV.

Quantum-chemical simulations can help predict morphology, and we have recently published several examples based on the combined use of experimental findings and first-principles calculations.[23,38] This provides a deeper knowledge of the electronic, structural, and energetic properties controlling the morphology and the transformation mechanisms of different metals and metal oxides. Based on the Wulff theorem, the equilibrium shapes for these systems were obtained from the values of their respective surface energies. These investigations are useful to gain further understanding on how to achieve morphological control of complex three-dimensional crystals by tuning the surface energy ratios for different facets. It is important to note that these theoretical results consider the crystals to be *in vacuum*. Therefore, in order to obtain the experimental morphology, we proposed the hypothesis where the interaction between the environments during the synthesis process could decrease/increase the surface energy of the exposed surfaces. Theoretically, the achieved morphology depends on the surface energy of each crystal termination.

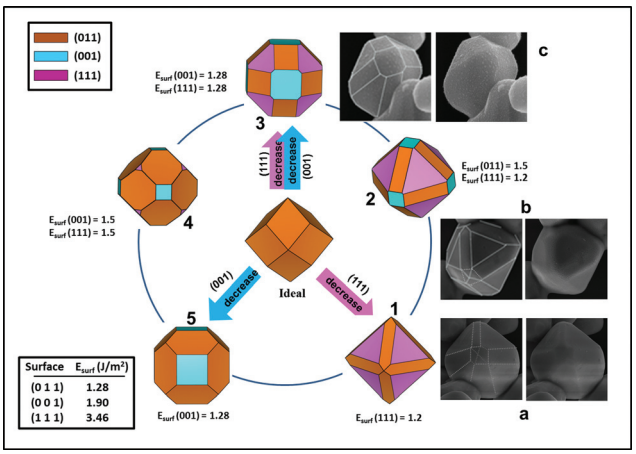

Fig. 2 Theoretical morphology derived from the Wulff construction and FE-SEM images of the β-Ag_2MoO_4 microcrystals obtained using (a) water, (b) ammonia, and (c) ethanol.

After the corresponding convergence test on the systems, slab models containing 6, 8 and 6 molecular units for (001), (011), and (111) surfaces were considered, with areas of 44.7, 63.2, and 38.7 Å^2, respectively. It is worth noting that (001) and (011) surfaces are O- and Ag-terminated, while the (111) surface is O- and Mo-terminated.[38] Analysis of Fig. 2 shows that the ideal morphology was dominated by the (011) orientation (in its three equivalent possibilities, as (011), (110), and (101) considering the cubic symmetry). When the surface energy ratio was varied by decreasing the (111) surface energy value to 1.2 J m^{-2} through a random perturbation, the results revealed that both (011) and (001) facets were disfavored, and morphology 1 was observed. Furthermore, if the value of the (011) facet increased to 1.5 J m^{-2}, the (001) facet appeared, and morphology 2 was obtained. The FE-SEM images of β-Ag_2MoO_4 microcrystals obtained using water and ammonia as solvents are included as Fig. 2(a) and (b), respectively, being both possible candidates to theoretical morphologies 1 and 2. On the other hand, when the values of the surface energy for both (001) and (111) facets decreased to 1.28 J m^{-2} (the same value as the (011) surface energy), morphology 3 was obtained, which could be compared to the experimental morphology of β-Ag_2MoO_4 using ethanol as the solvent (c). Therefore, it was revealed that the experimental morphology could be modulated by tuning the surface stability, which is dependent on the solvent. In this sense, the morphology can be modulated by changing the surface energies of the different exposed surface planes during the crystal growth process, thus modifying the properties of the material.

3.3 Raman spectroscopy

Raman spectroscopy can be used as a probe technique to investigate the degree of structural order–disorder at the short-range of the materials. Group theory considerations lead to five Raman-active modes for the spinel structure,[15] as shown in eqn (2).

$$\Gamma = A_{1g} + E_g + 3T_{2g} \tag{2}$$

Considering a strong Mo–O bonding within the [MoO_4] tetrahedral cluster, this irreducible representation is split into four so-called "internal modes" (A_{1g} + E_g + 2T_{2g}) derived from the four fundamental vibration modes ν_1, ν_2, ν_3, and ν_4 of the free tetrahedron, and one "external mode" T_{2g} related to a translation of the Ag cation.[39,40]

The MR spectra of β-Ag_2MoO_4 obtained using different solvents are depicted in Fig. 3, and the data are summarized in Table 1. These spectra reveal the presence of five Raman-active vibrational modes as expected. Our experimental Raman shifts are in good agreement with previously reported values.[14–16,23,26,41–45]

For direct spinel molybdates Na_2MoO_4 and/or Ag_2MoO_4, five Raman modes were determined based on relative intensities,[39] isotopical effects,[42] polarized Raman,[43] and theoretical approach using a lattice dynamic calculation.[15,44] In Table 1,

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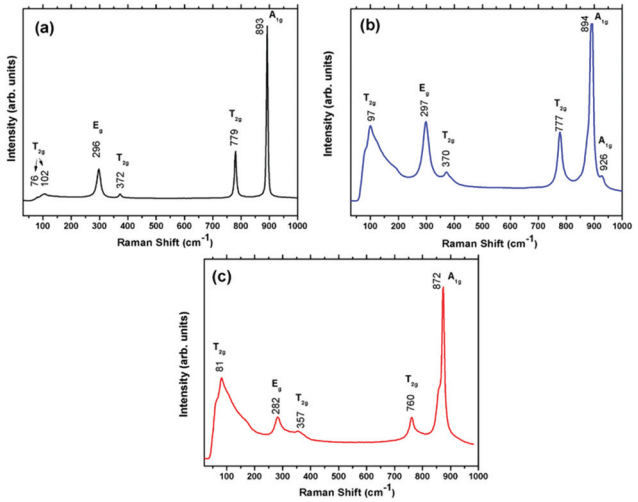

Fig. 3 MR spectra of the β-Ag_2MoO_4 microcrystals obtained using different solvents: (a) ethanol, (b) ammonia, and (c) water.

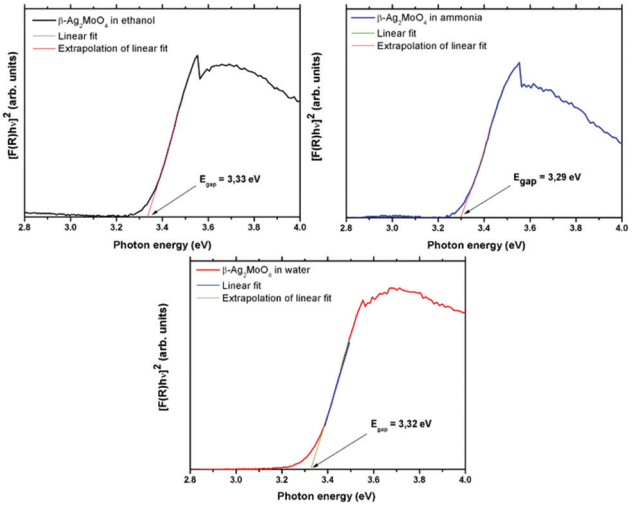

Fig. 4 UV-Vis diffuse reflectance spectra of the β-Ag_2MoO_4 microcrystals obtained using different solvents.

the experimental and theoretical values for the relative positions of the Raman-active modes are listed.

The T_{2g} "external mode" is difficult to observe,[15,26,45] being detected at 91 cm^{-1} as a single line in somewhat large crystals.[26] In our case, the external mode was split, a signature consistent with the removal of degeneracy. This suggests the creation of local defects affecting mainly the Ag cations in our samples. Hence, it seems reasonable to suppose that some Ag cations could be shifted from the center of their octahedral environment, especially when close to the nanocrystal surface.

3.4 Optical absorption

The UV-Vis diffuse reflectance spectrum of β-Ag_2MoO_4 synthesized using different solvents is reported in Fig. 4. The optical band-gap energies calculated using the Kubelka–Munk method[46] were 3.33, 3.32, and 3.29 eV for β-Ag_2MoO_4 synthesized using ethanol, water and ammonia, respectively. According to Li *et al.*,[29] silver molybdate exhibits an optical absorption spectrum governed by indirect electronic transitions. Data in the literature concerning the band-gap of molybdates with monovalent counter cations is scarce. In previous reports we revealed a direct band-gap of 3.15 eV for β-Ag_2MoO_4 samples prepared by CP (pH = 7; 70 °C),[20] and indirect band-gaps in the 3.24–3.3 eV range for samples obtained using a MAH method at temperatures ranging between 100 and 160 °C.[27] The value of 3.32 eV revealed in the present study is close to the indirect band-gap of 3.37 eV reported by Li *et al.*[29] These results indicate the existence of intermediary energy levels within the optical band-gap of the β-Ag_2MoO_4 microcrystals using different solvents, which can be associated with a structural disorder of the tetrahedral [MoO_4] and octahedral [AgO_6] clusters.

The band structure and the density of states (DOS) projected on atoms and orbitals for β-Ag_2MoO_4 are illustrated in Fig. 5. Theoretical calculations provide a band-gap value of 2.58 eV for β-Ag_2MoO_4, with an indirect transition from the valence band (VB) Γ point to the X point of the conduction band (CB). The direct band gap at the Γ point is 2.81 eV, slightly lower than the experimental measurements. The bottom of the VB is mainly formed by hybridization between Ag 4d and O 2p orbitals, while the CB is mainly formed by hybridization between Mo 4d and O 2p orbitals. For both transition metals $d_{x^2-y^2}$ and d_{z^2} states produce the main contribution.

The band structure for the surfaces was also computed, with the (001) surface presenting a band-gap value of 1.35 eV ($\Gamma - \Gamma$), while the other surfaces ((011) and (111)) had a conducting behavior. The band-gap is closely related to the structural distortions in the material as shown by MR spectroscopy. Both structural distortions and band-gap energies play an

Table 1 Experimental and theoretical values for the relative positions of the Raman-active modes

Symmetry	Experimental Raman shift (cm^{-1})	Theoretical Raman shift (cm^{-1})	Number	Description
A_{1g}	872–926	841–921 [15]	ν_1	Symmetric stretching MoO_4
T_{2g}	760–777	761–776 [15]	ν_3	Asymmetric stretching MoO_4
T_{2g}	357–372	348–370 [15,44]	ν_4	Asymmetric bending MoO_4
E_g	282–297	272–279 [15,44]	ν_2	Symmetric bending MoO_4
T_{2g}	76–102	89–93 [15]	—	External mode

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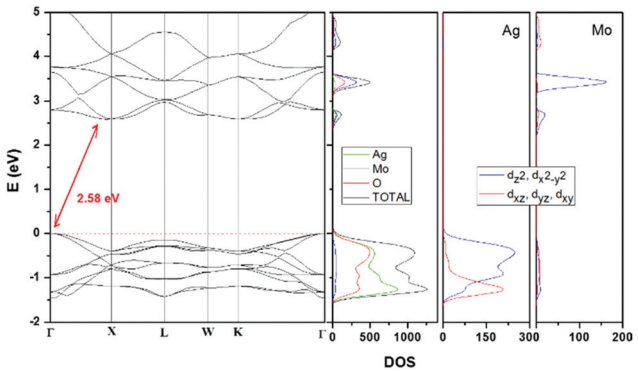

Fig. 5 Band structure and DOS projected on atoms and orbitals for β-Ag_2MoO_4.

important role in the electronic properties, such as photocatalytic activity and photoluminescence of the materials.

3.5 Photoluminescence

Fig. 6 shows the PL emission spectra of the β-Ag_2MoO_4 microcrystals synthesized using different solvents. These results indicate that the solvent affects the variation in the defect density of this material. The PL emission profile analysis shows a maximum emission in the blue region (449 nm) that is related to distortions in the tetrahedral [MoO_4] clusters. Furthermore, we can observe a weak emission in the orange (650 nm) region related to an ordered structure. The β-Ag_2MoO_4 microcrystal lattice can occur at different levels of distortion on the [AgO_6] and [MoO_4] complex clusters, which are responsible for PL emissions.

The cluster-to-cluster transfer charge (CCCT) in the β-Ag_2MoO_4 microcrystal containing more than one kind of cluster is described by excitations involving electronic transitions from one cluster to another cluster. Therefore, the main difference in PL spectra of β-Ag_2MoO_4 microcrystals can be attributed to the ratio between ordered $[MoO_4]_o/[AgO_6]_o$ and disordered $[MoO_4]_d/[AgO_6]_d$ clusters.

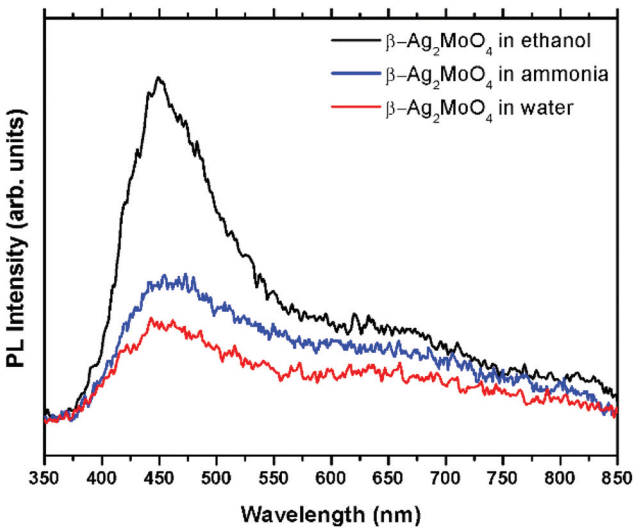

Fig. 6 PL emission spectra recorded at room temperature of the β-Ag_2MoO_4 microcrystals synthesized using different solvents.

The order–disorder effect is produced by the solvent, which induces the formation of different permanent dipoles, as a consequence of which the crystal has different polarization. Thus, the CCCT leads to the formation of different levels in the forbidden band gap region. This phenomenon of order/disorder has its origin in the crystal organization, which undergoes variation depending on the lattice interactions [MoO_4] cluster and modifier of the [AgO_6] cluster. The order–disorder effect in the lattice constant promotes the electronic charge transfer between the ordered and disordered, which is possible according to the following equations:

$$[MoO_4]_o^x + [MoO_4]_d^x \rightarrow [MoO_4]'_o + [MoO_4]_d^{\bullet} \quad (3)$$

$$[AgO_6]_o^x + [AgO_6]_d^x \rightarrow [AgO_6]'_o + [AgO_6]_d^{\bullet} \quad (4)$$

In these equations, the superscript x indicates neutral clusters, (′) the cluster with one electron, and (•) the cluster with one hole. These complex clusters induce shallow or deep defects that can modify the properties of β-Ag_2MoO_4. In this case, the PL properties observed are different, so these clusters have complex stability of electron–hole more or less effective interaction, because they are on different excited states. The electron–hole pairs are trapped by the complex mode different clusters, namely clusters that trap spin electrons as the same are more stable because they need to replace the spin interacting with each other.

3.6 Antifungal effect

The results demonstrated that, regardless of the solvent used in the synthesis, the β-Ag_2MoO_4 microcrystals had both fungistatic and fungicidal activity against *C. albicans*. In addition, for each of the three samples obtained, that is, β-Ag_2MoO_4 microcrystals synthesized using water, ethanol, and ammonia as solvents, the concentration that resulted in no visible growth by visual inspection (MIC value) also produced no colonies on the plates after incubation (MFC value). However, when ethanol and ammonia were used as solvents, the MIC/MFC values were similar (7.81 μg mL^{-1}) and lower than that observed for the sample synthesized using water (15.62 μg mL^{-1}) (Fig. 7). It is worth noting that, even at concentrations lower than MIC/MFC values, the β-Ag_2MoO_4 microcrystals were still able to reduce yeast growth. When *C. albicans* was incubated without β-Ag_2MoO_4 microcrystals (control), a count of 6.6 $\log_{10}$ CFU mL^{-1} was obtained. At half of the MIC/MFC value (3.90 mg mL^{-1}), yeast growth was also reduced to approximately 3.5 $\log_{10}$ CFU mL^{-1} and 2.3 $\log_{10}$ CFU mL^{-1}, for β-Ag_2MoO_4 obtained using ammonia and ethanol as solvents, respectively. The effect of the β-Ag_2MoO_4 microcrystals synthesized using water was less pronounced, and a reduction of the $\log_{10}$ CFU mL^{-1} value compared to the control (from 6.6 $\log_{10}$ CFU mL^{-1} to 3.0 $\log_{10}$ CFU mL^{-1}) was only observed at a concentration of 7.81 μg mL^{-1} (half of MIC/MFC value). These results demonstrate that the synthesized β-Ag_2MoO_4

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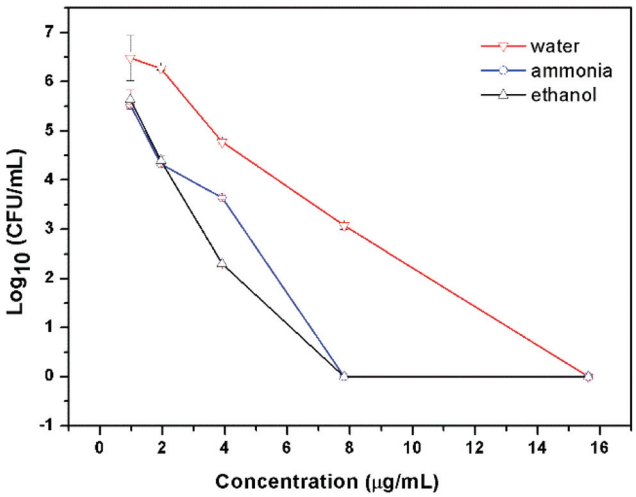

Fig. 7 Representation of $\log_{10}$ CFU mL^{-1} *versus* the sub-inhibitory concentrations of the β-Ag_2MoO_4 synthesized using different solvents. Control = 6.6 $\log_{10}$ CFU mL^{-1}.

microcrystals have great potential as antifungal agents, particularly when ethanol and ammonia are used as solvents during the synthesis.

To rationalize the fungistatic and fungicidal activity, physical mechanisms were based on the interaction between the *C. albicans* and the nanoscale surface structure. As shown in the simulated single crystals and FE-SEM images (Fig. 2), the morphologies of ethanol and ammonia prepared materials have larger contribution of the surface planes (001) and (011) compared to that of the water, which are facets with exposed Ag and O atoms to the vacuum. This would suggest that silver ions would be more readily available to diffuse/dissolve into the media thereby eliciting the enhanced potency due to a higher concentration of silver ions. These results indicate that the β-Ag_2MoO_4 microcrystals with superficial/structural defects, which are caused by distorted $[MoO_4]$/$[AgO_6]$ clusters, are effective in the inactivation of *C. albicans* fungus. An explanation for this inactivation or destruction is related to the presence of specific defects on the crystal surface, that can polarize the lattice and lead to electronic transitions between disordered/distorted $[MoO_4]^{\bullet}_{d}$ and $[AgO_6]^{\bullet}_{d}$ and ordered/ordered $[MoO_4]'_{o}$ and $[AgO_6]'_{o}$ clusters, which also produce OH* and O_2H* radicals. These oxidizing species inhibit fungus growth, and eventually result in fungus death. Fig. 8 shows the plausible mechanism for the inactivation of *C. albicans* fungus with β-Ag_2MoO_4 microcrystals.

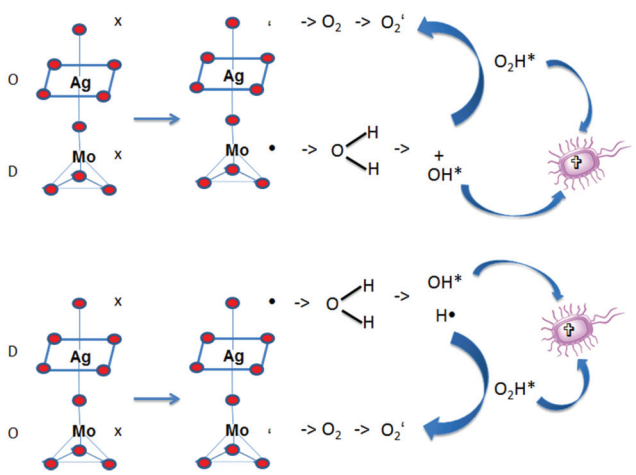

Fig. 8 Action mechanism of the β-Ag_2MoO_4 microcrystals.

4. Conclusions

In summary, β-Ag_2MoO_4 microcrystals were synthesized by a CP method using different solvents (water, ethanol and ammonia). The optical and antifungal properties of the β-Ag_2MoO_4 samples were studied by combining experimental techniques and first-principles calculations. We showed that β-Ag_2MoO_4 microcrystals exhibited faceted morphologies similar to those calculated by the Wulff construction. The predominant exposed facets are composed of (001), (011), and (111) surfaces, where β-Ag_2MoO_4 microcrystals with predominant (111) and (001) surfaces with low surface energy had superior antifungal activity for the inactivation of *C. albicans* fungus. The presence of intermediate energy levels within the optical band-gap can be associated with the structural disorder of the tetrahedral $[MoO_4]$ and octahedral $[AgO_6]$ clusters, which enhance the density of electron–hole pairs. The PL emissions depended critically on the structural disorder of these clusters. Furthermore, a theoretical model based on different β-Ag_2MoO_4 clusters was proposed to explain the underlying reason for the fungistatic and fungicidal activity against *C. albicans*. Our model also reveals some details of the influence mechanism, and gives some important information about how to improve the bactericidal properties through designing the morphology of the patterned surface.

Author contribution

Maria T. Fabbro and Luis P. S. Santos prepared the samples and performed the structural analysis and the UV–vis, micro-Raman (MR), FE-SEM images and photoluminescence measurements. André Perrin and Christiane Perrin interpreted MR. Camila C. Foggi, Carlos E. Vergani and Ana L. Machado performed the antifungal activity measurements. Lourdes Gracia, Eloisa Cordoncillo and Juan Andrés performed the theoretical calculations. Elson Longo conceived the project. All authors participated in writing the manuscript and discussion of the results.

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